Mock Exam

Methods and Databases in Empirical Finance

First Name, Surname: _____

Study Program: _____

Signature: _____

Student-ID: 0 __ __ __ __ __ __ __

0	☒	☐	☐	☐	☐	☐	☐	☐
1	☐	☐	☐	☐	☐	☐	☐	☐
2	☐	☐	☐	☐	☐	☐	☐	☐
3	☐	☐	☐	☐	☐	☐	☐	☐
4	☐	☐	☐	☐	☐	☐	☐	☐
5	☐	☐	☐	☐	☐	☐	☐	☐
6	☐	☐	☐	☐	☐	☐	☐	☐
7	☐	☐	☐	☐	☐	☐	☐	☐
8	☐	☐	☐	☐	☐	☐	☐	☐
9	☐	☐	☐	☐	☐	☐	☐	☐

Instructions:

- Total exam duration: **45 minutes (no extra time); (90 minutes in final exam)**
- Total points: **45** (it should take about one minute to complete one point); (90 points in final exam)
- Number of questions: There will be **14 questions** in this exam.
- **Provide your answers below the questions in the black boxes. Solutions written on external sheets will not be considered.**
- **No** negative points will be assigned.
- Do not unstaple the question sheets or detach pages from this exam. Otherwise, your exam will not be evaluated.
- Please do not round intermediate results.
- Throughout the exam, commas are used to separate thousands and dots are used to separate decimals, i.e. 10,000 EUR means ten thousand euros and 10.40 EUR means ten euros and forty cents.
- Please verify the completeness of your exam.
- Fill out the title page and the Student ID on **EVERY page. You will receive additional 2 minutes for this at the beginning of the exam.**
- If you need extra space, use the backsides of each page.
- Only the following resources are allowed to be used:
  - Dark pen,
  - Non-electronic dictionary,
  - Non-graphical and non-programmable calculator.

Good luck!

Part 1: Open Questions

Question 1

You are given the following R-Script to handle CRSP data. Discuss the data filters applied in this setting.

```
crsp_monthly <- msf_db |>
  filter(mthcaldt >= start_date & mthcaldt <= end_date) |>
  select(-c(siccd, primaryexch, conditionaltype, tradingstatusflg)) |>
  inner_join(
    stksecurityinfohist_db |>
      filter(sharetype == "NS" &
               securitytype == "EQTY" &
               securitysubtype == "COM" &
               usincflg == "Y" &
               issuertype %in% c("ACOR", "CORP") &
               conditionaltype %in% c("RW", "NW") &
               tradingstatusflg == "A") |>
      select(permno, secinfostartdt, secinfoenddt,
             primaryexch, siccd),
    join_by(permno)
  ) |>
  filter(mthcaldt >= secinfostartdt & mthcaldt <= secinfoenddt) |>
  mutate(date = floor_date(mthcaldt, "month")) |>
  collect() |>
  mutate(
    date = ymd(date),
    shrout = shrout * 1000
  )
```

maybe there is a mistake, or sth missing -> discuss,
what does a sorting look like? (tensile, etc.)
breakpoints

(6 points)

NS = common shares,
EQTY = Equity Shares,
COM = Common stocks from Equity,
Y = Only US companies, even if international companies in the US exchange,
ACOR = Active corporations,
CORP = Legacy corporations exclusing REITS,
RW, NW = Real world trading securities,
A = Actively trading securities

Question 2

Based on the information provided below, calculate the expected CAPM return for firm A and B assuming $\alpha = 0$ and a market excess return of 2%.

Sample Variance–Covariance Matrix (A, B, C, D, Market)

	A	B	C	D	Market
A	0.0481	0.0352	0.0192	0.0416	0.0320
B	0.0352	0.0709	0.0264	0.0572	0.0440
C	0.0192	0.0264	0.0369	0.0312	0.0240
D	0.0416	0.0572	0.0312	0.0901	0.0520
Market	0.0320	0.0440	0.0240	0.0520	0.0625

(4 points)

B(A) = 0.0320/0.0625 = 0.512
CAPM return = 0.512 (2) = 1.024

Question 3

left side = predicted
right side = expected

Consider the linear factor model

$$\hat{r}_i = \alpha + \beta_1 \text{Beta}_i + \beta_2 \text{Size}_i + \beta_3 \text{Value}_i + \beta_4 \text{Profitability}_i,$$

with

$$\alpha = 0.015,\ \beta_1 = 0.045,\ \beta_2 = 0.100,\ \beta_3 = -0.020,\ \beta_4 = 0.060.$$

A firm has the following characteristics:

$$\text{Beta} = 1.4,\ \text{Size} = 0.6,\ \text{Value} = 0.5,\ \text{Profitability} = 0.10.$$

1. Compute the predicted monthly return $\hat{r}$. plug and play
2. If the realized (actual) return is $r = 0.05$, compute the residual. residual = diff. between predicted and actual
3. Suppose the realized return equals the predicted return. What does this imply about how well the model explains this firm's return in that month?

(3 points)

Question 4

Name three key differences between cross-sectional asset pricing and empirical corporate finance, focusing on their research objectives, primary variables of interest, and empirical approaches.

(3 points)

1. what qs they are trying to ans.
2.

Question 5

Outline the difference between Primary and Secondary Markets.

(3 points)

Question 6

Explain the concept of the Efficient Frontier and the Capital Market Line graphically and briefly describe its components.

(6 points)

Question 7

Name and explain three behavioral biases discussed in the lecture. In your answer, clearly underline the name of each bias and provide a brief example illustrating how each bias can affect investor decision-making.

(3 points)

1. over confidence
2. herding
3. home bias
4. ...

Question 8

Explain the Fama-MacBeth regression methodology used in empirical asset pricing. Describe the steps involved, including what is estimated in each step.

(4 points)

1. cross-sectional
2. estimating

both steps, outputs for each step

Question 9

Explain the concept of P-hacking in finance. Give two examples how concerns about P-hacking could be mitigated.

(3 points)

running a large amount of tests, and one of them will become sig. = and choosing this

Part 2: Single Choice Questions

Question 10

What is the fundamental capital allocation problem that capital markets are designed to solve?

- ☐ A) Capital is scarce relative to investment opportunities.
- ☐ B) Those who have capital are typically the same as those who need capital.
- ☐ C) Those who have capital are not the ones who need capital.
- ☐ D) Capital markets exist primarily to reduce investment risk.

(2 points)

Question 11

Why is total return preferred over price return in empirical finance studies?

- ☐ A) Total return excludes dividends to reduce noise.
- ☐ B) Price return overstates performance for growth stocks.
- ☐ C) Total return captures both capital gains and dividend income.
- ☐ D) Price return cannot be compared across firms.

(2 points)

Question 12

What is the key economic interpretation of the Capital Market Line (CML)?

- ☐ A) It shows the relationship between firm size and expected returns.
- ☐ B) It represents the optimal trade-off between risk and return using the risk-free asset and the tangency portfolio.
- ☐ C) It explains how individual stock returns depend on firm-specific risk.
- ☐ D) It applies only to inefficient portfolios below the efficient frontier.

(2 points)

Question 13

Which type of risk is rewarded with a risk premium according to the CAPM?

- ☐ A) Idiosyncratic risk that is unique to a firm.
- ☐ B) Total risk measured by return variance.
- ☐ C) Systematic risk that cannot be diversified away.
- ☐ D) Liquidity risk arising from trading frictions.

(2 points)

Question 14

What is the main conclusion of Fama and French (2010) regarding mutual fund performance?

- ☐ A) Most active funds outperform the market before and after fees.
- ☐ B) Aggregate mutual fund alpha is close to zero before fees and negative after fees.
- ☐ C) Skilled managers consistently deliver positive net alpha.
- ☐ D) Performance persistence is strong across the entire fund distribution.

(2 points)

End of the exam!

1. factor model (CAPM most probably), understand FF3,FF5 (basics)
2. Beta, min var. potfolio, optimal port. (formulas)
3. efficient frontier
4. MSE formula
5. papers: 3-5 slides discussed, how is the model built, etc. (P-hacking, Huert paper - microcaps be tested again, Stein - model, Fama) similar patterns
6. reading assignment paper (2 - uploaded)
7.